



Cloud Computing
Assignment-Week 2
TYPE OF QUESTION: MCQ/MSQ

Number of questions: 10

Total mark: 10 X 1 = 10

QUESTION 1:

A virtualization solution requires the guest operating system to invoke **hypercalls** instead of executing certain privileged instructions directly. This approach is known as:

- A. Full Virtualization
- B. Hardware-assisted Virtualization
- C. Para-virtualization
- D. Containerization

Correct Answer: C

Detailed Solution:

Para-virtualization modifies the guest operating system so that privileged operations are performed through hypercalls to the hypervisor. This reduces virtualization overhead but requires OS modification.

QUESTION 2:

Which of the following is a key characteristic of full (hardware-assisted) virtualization, as opposed to para-virtualization?

- A. Guest OS must be modified to issue hypercalls to the hypervisor.
- B. Guest OS runs unmodified, with the hypervisor and CPU extensions handling privileged instructions.
- C. Guest and host must share the same operating system kernel.
- D. No hypervisor is involved at all.

Correct Option: B

Detailed Answer: In full (hardware-assisted) virtualization, an unmodified guest OS runs while CPU extensions (e.g., Intel VT-x, AMD-V) and the hypervisor transparently handle privileged instructions. Option A describes para-virtualization (guest OS modified via hypercalls), C describes containers, and D is false since a hypervisor is always present.

QUESTION 3:

Which statement about a public cloud is MOST accurate?

- A. It is dedicated to, and controlled by, a single organization behind its own firewall.



- B. Its infrastructure is provisioned for open use by the general public and is owned and operated by a cloud service provider.
- C. It is shared only among organizations that have common regulatory concerns.
- D. It always guarantees stronger data isolation than a private cloud.

Correct Option: B

Detailed Answer: A public cloud is provisioned for open use by the general public and is owned, managed, and operated by a cloud service provider on its own premises. Option A describes a private cloud, C describes a community cloud, and D is false since private clouds typically provide stronger isolation.

QUESTION 4:

An application must repeatedly navigate, randomly access, and modify elements of a moderately sized XML document held in memory. Which XML parser API is MOST appropriate?

- A. SAX (push-based, event-driven, forward-only)
- B. StAX (pull-based, event-driven, forward-only)
- C. DOM (builds an in-memory tree allowing random access and modification)
- D. None; an XML document cannot be modified once parsed

Correct Option: C

Detailed Answer: DOM builds a complete in-memory tree representation of the document, enabling random access, navigation, and in-place modification of elements. This suits documents that fit in memory and require manipulation. SAX and StAX are forward-only streaming parsers not designed for random access or modification, so they are unsuitable here.

QUESTION 5:

Which statement correctly distinguishes a 'host OS', a 'guest OS', and a 'hypervisor'?

- A. The host OS runs inside a VM, the guest OS runs on bare metal, and the hypervisor is a user application.
- B. The hypervisor creates and manages VMs; each guest OS runs inside a VM; and for a Type-2 hypervisor, the host OS is the OS on which the hypervisor runs.
- C. The guest OS and host OS are always identical copies of the same operating system.
- D. The hypervisor is simply the hardware layer located beneath the host OS.

Correct Option: B

Detailed Answer: The hypervisor creates and manages virtual machines; each guest OS runs inside its own VM; and for a Type-2 (hosted) hypervisor, the host OS is the operating system on



which the hypervisor runs as an application. Option A reverses the host/guest roles, C is false, and D incorrectly describes the hypervisor as hardware.

QUESTION 6:

An enterprise migrates its private cloud to a third-party managed data center. Which of the following concerns should be evaluated?

- (i) Service availability guarantees (SLAs)
- (ii) Compliance with data protection regulations
- (iii) Network latency between users and cloud resources

- A. Only (i)
- B. (i) and (ii)
- C. (ii) and (iii)
- D. (i), (ii), and (iii)

Correct Answer: D

Detailed Solution:

Hosted private clouds require organizations to evaluate SLA commitments, legal compliance regarding data location, and network performance because applications now depend on external connectivity.

QUESTION 7:

In the traditional web services stack, which technology is used to describe a web service's interface — its operations, input/output messages, and how to invoke it?

- A. UDDI
- B. WSDL
- C. SMTP
- D. HTML

Correct Option: B

Detailed Answer: WSDL is an XML-based language that describes a web service's interface: its operations, message formats, and binding/invoke details. UDDI is the registry used for discovery, SMTP is an email transport protocol, and HTML is a markup language for web pages.



QUESTION 8:

Traditional SOAP-based web services provide which of the following capabilities?

- (i) Platform-independent communication
- (ii) XML-based message exchange
- (iii) Automatic service discovery through UDDI
- (iv) Guaranteed microsecond-level response latency

- A. (i), (ii), and (iii)
- B. (ii) and (iv)
- C. Only (i)
- D. All of the above

Correct Answer: A

Detailed Solution:

SOAP web services promote interoperability using XML and WSDL and support service discovery through UDDI. They do not guarantee extremely low-latency communication because XML processing introduces overhead.

QUESTION 9:

Which of the following problems are effectively addressed by traditional web services, and which are NOT?

- (i) Passing through firewalls using standard HTTP/HTTPS
- (ii) Enabling interoperability across different programming languages and operating systems
- (iii) Ultra-low-latency, real-time inter-service communication
- (iv) Service discovery through registries like UDDI

- A. Only (i) and (ii) are addressed
- B. (i), (ii), and (iv) are addressed; (iii) is NOT addressed
- C. Only (ii) is addressed
- D. All are addressed

Correct Option: B

Detailed Answer: Traditional web services address firewall transparency by using HTTP/HTTPS, ensure platform and language interoperability via XML, SOAP, and WSDL, and support service



discovery through registries like UDDI. However, they do not provide ultra-low-latency, real-time communication, as XML parsing and SOAP overhead introduce delays, making modern alternatives like gRPC better suited for real-time scenarios.

QUESTION 10:

In the Service-Oriented Architecture (SOA) model, which set of operations correctly maps the interactions among the service provider, service registry, and service requestor?

- A. Publish (provider → registry), Find (requestor → registry), Bind (requestor → provider)
- B. Publish (requestor → provider), Find (provider → registry), Bind (registry → requestor)
- C. Bind (provider → registry), Publish (requestor → registry), Find (provider → requestor)
- D. Find (provider → registry), Publish (requestor → registry), Bind (provider → provider)

Correct Option: A

Detailed Answer: In SOA, the provider publishes its service description to the registry, the requestor finds a suitable service by querying the registry, and the requestor then binds to and invokes the provider directly. This publish–find–bind triangle defines the interactions among the three roles.
